

Differentiating control flow



A strategy for **If**, **Scan** and **Loop**

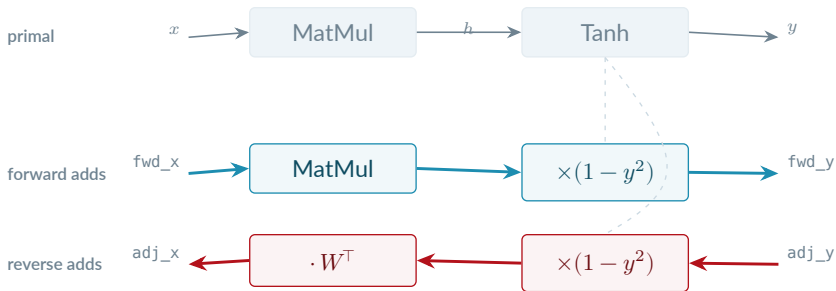
onnx-ad 0.1.0

design brief

onnx-ad

22 September 2026

One graph, three readings



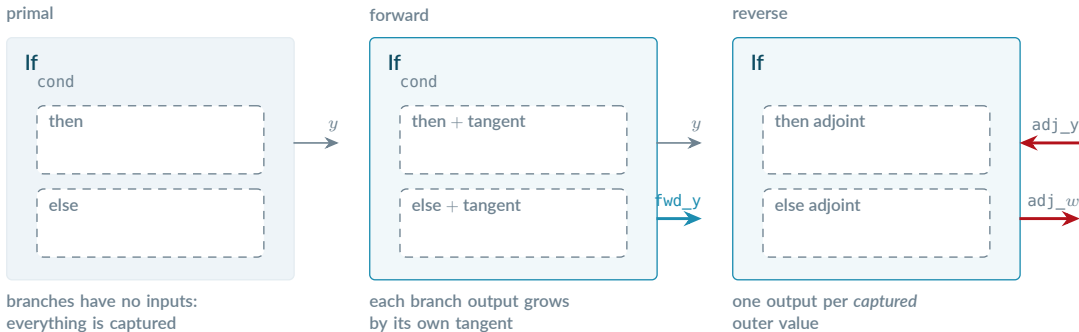
A pass *adds* a band and keeps the primal, so a nonlinear rule reads the tensor the primal already produced – the dashed links. Everything after this slide is the same picture with a subgraph inside a node.

Why If, then Scan, then Loop

If	new capability needed recursion into subgraphs, outer-scope capture	reverse mode needs nothing iterative
Scan	a tape, and a sweep in the opposite direction	a trip count known before the primal runs
Loop	a trip count known only after the primal runs	the Scan machinery, fed by that trip count

The adjoint of a Loop is a Scan: once the primal loop has run, its trip count is data.

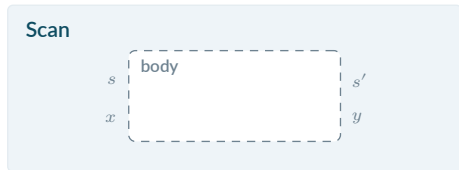
If – differentiate inside the branch



Only the taken branch computes its derivative – which is why not to evaluate both and select with **Where**.

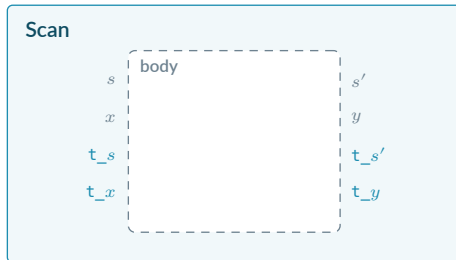
Scan forward – one Scan, no tape

primal



$\text{num_scan_inputs} = M$

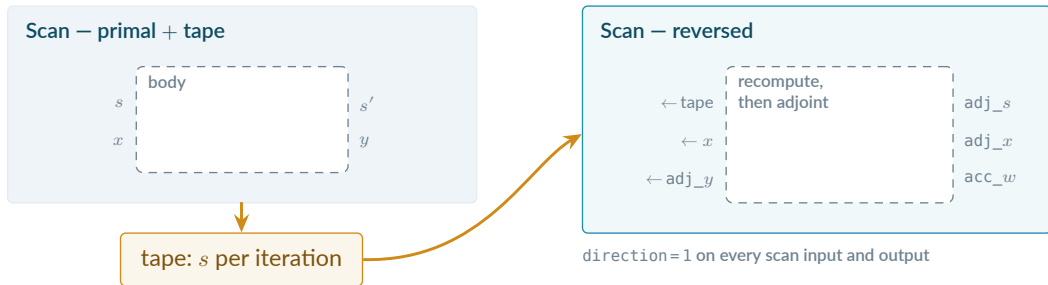
forward



$\text{num_scan_inputs} = 2M$

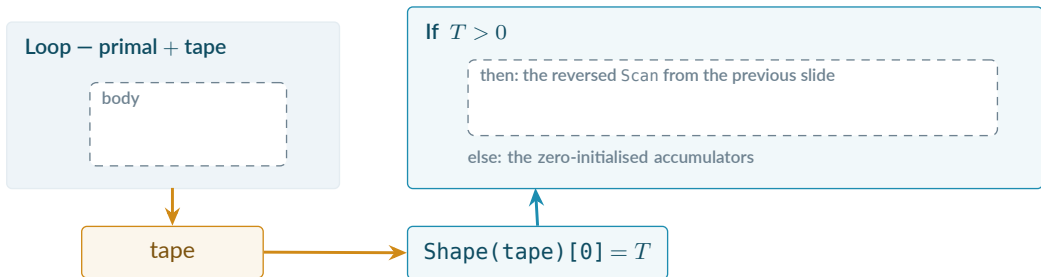
The tangent of a scan is a scan of the same length whose state is the pair $(s, \dot{s})$: two× the body, any number of seeds, nothing stored.

Scan reverse – tape the state, sweep backwards



The taped input state is the minimal recomputation root – and a captured weight accumulates in the *state*, so its adjoint costs $O(|w|)$, not $O(T \cdot |w|)$.

Loop reverse – the trip count becomes data



A reversed Loop tolerates $T = 0$ natively, but needs `Gather(tape, T-1-i)` and a final flip of its scan outputs.

The reverse sweep is a Scan, not a Loop: reading backwards removes all index arithmetic.

Checked against the runtime, and the order of work

ONNX Runtime 1.30 — measured, not assumed

- ✓ outer-scope capture, two levels deep
- ✓ reversed scan input *and* output directions
- ✓ a scan state carrying a trailing seed axis
- ✓ loop trip count from `Shape(tape)[0]`
- ✗ a zero-length Scan is rejected
- ✓ an If guarding that empty case

staging

- 0 unroll for a static trip count — small, useful alone, and the test oracle
- 1 scope chain, shared name allocator, capture analysis; then If
- 2 Scan forward, then Scan reverse
- 3 Loop, on top of Scan's

Unrolling is the oracle: two Jacobians from two independent implementations, needing no framework to referee them.